Modeling Auto Insurance Claim Probability Using Supervised and Unsupervised Learning

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I. Introduction

Road accidents and vehicle incidents cost billions in insurance claims annually. Accurate insurance claim prediction is critical for insurers to price premiums based on actual risk [1]. It also benefits the customer because low risk drivers shouldn't have to pay for high risk drivers. Furthermore, customers pay what you actually owe instead of an overestimate. This project is divided into two parts. The first part is a supervised learning model for predicting the likelihood that a customer files an auto insurance claim within the next year. We apply and compare three classification models: Logistic Regression, Random Forest, and Multi-Layer Perceptron (MLP). Each of these classifying algorithms will be evaluated on accuracy, F1-score, and ROC-AUC. The second part involves Clustering. K-means is applied to identify natural risk segments within the policyholder population, providing interpretable groupings that complement the supervised predictions. The importance is to understand trends on what type of people are more likely to file claims. This allows companies to better price and accommodate. The dataset used is a publicly available Kaggle dataset titled "Customer Data" [2], consisting of 10,000 policyholder records.

II. Dataset & Features

A. Preprocessing

Some columns were removed prior to modeling. The `id` column was dropped as it is a unique row identifier with no pattern. The `postal_code` column was removed because raw postal codes are not geographically meaningful. The same code can span multiple locations. The `race` column was dropped for two reasons: the class distribution was severely imbalanced (approximately 90% majority, 10% minority), and using race as a predictive feature for insurance pricing is both ethically problematic and legally prohibited in many jurisdictions.

Two features contained missing values: `credit_score` (982 missing, 9.8%) and `annual_mileage` (957 missing, 9.6%). Dropping all affected rows would have eliminated approximately 18.5% of the dataset, which is too significant. Median imputation was applied instead, as the median is robust to the outliers observed in both distributions.

B. Feature Engineering

Ordinal encoding was applied to features that can be naturally ranked: `age`, `driving_experience`, `education`, and `income`. Each was mapped to integer ranks preserving their inherent order.

Binary encoding converted two-class categorical variables: `gender`, `vehicle_year`, `vehicle_ownership`, `married`, and `children` to 0/1 values.

An additional one-hot encoding was applied to `vehicle_type`, which contains no natural ordering between sedan and sports car, producing a single binary indicator column `is_sports_car`.

The dataset was split 80/20 into training and test sets using stratified sampling to preserve the original class distribution (68.7% no claim, 31.3% claim) across both splits, as shown in Fig. 1. Given the mild class imbalance, `class_weight='balanced'` was applied to all classifiers rather than resampling techniques such as SMOTE, as the imbalance was not severe enough to warrant synthetic data generation.

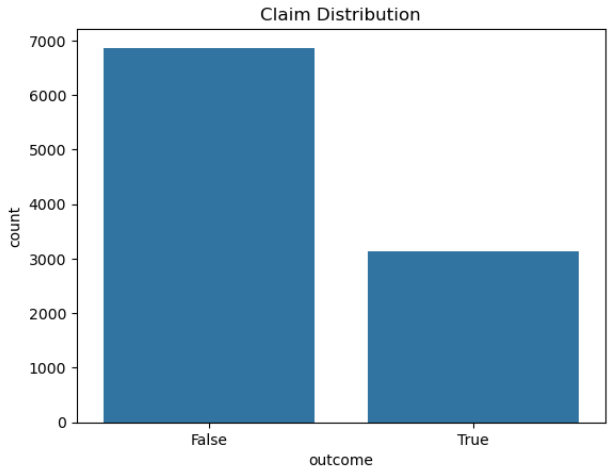

Fig. 1. Class distribution of the outcome variable. 68.7% of policyholders did not file a claim (False) while 31.3% did (True), indicating a mild class imbalance.

Fig. 2 shows feature distributions by claim outcome. Counterintuitively, non-claimants exhibit higher average speeding violations (1.925 vs 0.514), DUIs (0.310 vs 0.084), and past accidents (1.404 vs 0.294). This suggests claimants tend to report incidents more readily rather than exhibiting inherently riskier behavior.

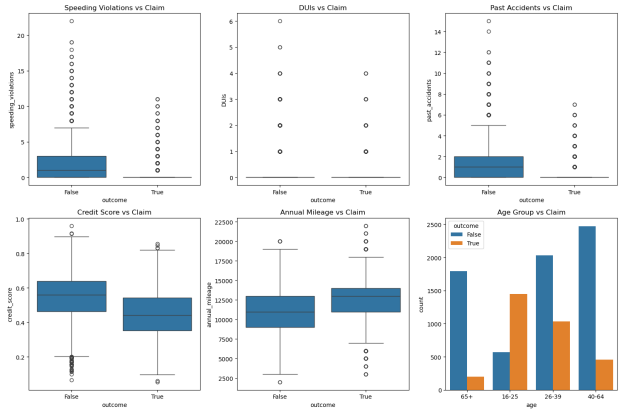

Fig. 2. Feature distributions by claim outcome. Non-claimants have higher speeding violations, DUIs, and past accidents on average. A counterintuitive finding suggesting claimants report incidents more readily than non-claimants.

C. Feature Scaling

StandardScaler was applied to all features, transforming each to zero mean and unit variance:

$$x' = \frac{x - \mu}{\sigma} \tag{1}$$

where x is the original feature value, μ is the feature mean, σ is the feature standard deviation, and x' is the resulting scaled value. This was done before all 3 models for controlled variable consistency, however it was only required for MLP.

MLP training relies on gradient descent to minimize a loss function $L(w)$, where $w \in \mathbb{R}^d$ is the vector of model weights and d is the number of parameters. The weight update rule at each iteration is:

$$w \leftarrow w - \eta \frac{\partial L}{\partial w} \tag{2}$$

where $\eta > 0$ is the learning rate and $\frac{\partial L}{\partial w}$ is the gradient of the loss with respect to the weights. Scaling ensures all gradients are on the same order of magnitude, producing a well-conditioned loss surface and stable convergence.

Logistic Regression minimizes the log-loss:

$$L(w) = -\frac{1}{n} \sum_{i=1}^{n} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{3}$$

where n is the number of training samples, $y_i \in \{0, 1\}$ is the true label for sample i, and $\hat{y}_i = \sigma(w^\top x_i) \in (0, 1)$ is the predicted probability produced by the sigmoid function $\sigma(\cdot)$. While Logistic Regression is less sensitive to scale than MLP, iterative solvers such as `liblinear` and `lbfgs` rely on gradient-based optimization and converge faster with scaled inputs. Without scaling, features with larger magnitudes produce disproportionately large weights, distorting the L2 regularization penalty formulated in Section III-D.

Random Forest makes decisions by finding the optimal split threshold t for each feature x_j at each node:

$$\text{split: } x_j \leq t \tag{4}$$

The quality of a split is measured by the reduction in Gini impurity G:

$$G = 1 - \sum_{k=1}^{K} p_k^2 \tag{5}$$

where p_k is the proportion of samples belonging to class $k \in \{0, 1\}$ at that node. Since Gini impurity depends only on relative ordering, not absolute magnitude, Random Forest is theoretically invariant to scaling. Therefore Random Forest is theoretically invariant to feature scaling. Scaling was applied uniformly across all three models solely to ensure a controlled experimental comparison.

III. Methodology

Three supervised classification models were implemented and compared, followed by K-Means clustering for unsupervised risk segmentation. All classification models were trained on the same scaled training set to ensure a controlled comparison.

A. Logistic Regression

Logistic Regression serves as the baseline model. It models the probability of a binary outcome as:

$$\hat{y} = \sigma(w^\top x + b) = \frac{1}{1 + e^{-(w^\top x + b)}} \tag{6}$$

where $w \in \mathbb{R}^d$ is the weight vector, $x \in \mathbb{R}^d$ is the input feature vector, b is the bias term, and $\sigma(\cdot)$ is the sigmoid function mapping outputs to $(0, 1)$. It was chosen as a baseline due to its interpretability and computational efficiency. `class_weight='balanced'` was set to account for the mild class imbalance (68.7% no claim, 31.3% claim), ensuring the model does not bias toward the majority class.

The model applies L2 regularization, adding the following penalty to the log-loss:

$$R(w) = \lambda \|w\|^2 = \lambda \sum_{j=1}^{d} w_j^2 \tag{7}$$

where $\lambda = \frac{1}{C}$ and $C > 0$ is the inverse regularization hyperparameter. L2 was chosen over L1 because all 15 features retained after preprocessing are plausibly relevant to claim prediction. L2 shrinks weights toward zero without eliminating any feature entirely, which is appropriate here. Hyperparameters tuned are discussed in Section III-D.

B. Random Forest

Random Forest is an ensemble method that builds T independent decision trees and aggregates their predictions by majority vote:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \ldots, h_T(x)\} \tag{8}$$

where $h_t(x)$ is the prediction of the t-th tree. Each tree is trained on a bootstrap sample of the training data with a random subset of features considered at each split, reducing variance through decorrelation of the individual trees. `class_weight='balanced'` was applied. Hyperparameters tuned are discussed in Section III-D.

C. Multi-Layer Perceptron (MLP)

The MLP is a feedforward neural network that learns non-linear decision boundaries through successive layers of transformations:

$$h^{(l)} = \text{ReLU}(W^{(l)}h^{(l-1)} + b^{(l)}) \tag{9}$$

where $h^{(l)}$ is the activation vector at layer l, $W^{(l)}$ is the weight matrix, $b^{(l)}$ is the bias vector, and $\text{ReLU}(z) = \max(0, z)$ is the activation function introducing non-linearity. The output layer applies a sigmoid activation for binary classification. Early stopping with `validation_fraction = 0.1` was applied so 10% of training data is held out as a validation set and training halts when validation performance stops improving, preventing overfitting. Hyperparameters tuned are discussed in Section III-D.

D. Hyperparameter Tuning

All three models were tuned using RandomizedSearchCV with 5-fold cross-validation, optimizing for ROC-AUC. The ROC-AUC score is averaged across all 5 folds to produce a robust estimate of generalization performance. Rather than exhaustively evaluating all parameter combinations as in GridSearchCV, RandomizedSearchCV samples `n_iter = 20` random combinations from the parameter space, providing a computationally efficient approximation. The best estimator from each search was retained for final test set evaluation.

Logistic Regression:

- $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$: Regularization strength; small C underfits, large C overfits.
- Solver $\in$ {`lbfgs`, `liblinear`}: Optimization algorithm; `liblinear` suited for smaller datasets.
- `max_iter` $\in \{100, 200, 500\}$: Maximum solver iterations.

Random Forest:

- `n_estimators` $\in \{50, 100, 200, 300\}$: Number of trees; more trees reduce variance.
- `max_depth` $\in \{\text{None}, 5, 10, 20\}$: Tree depth; constraining acts as regularization.
- `min_samples_split` $\in \{2, 5, 10\}$: Minimum samples required to split a node.
- `min_samples_leaf` $\in \{1, 2, 4\}$: Minimum samples at a leaf; higher values smooth the model.

MLP:

- `hidden_layer_sizes` $\in \{(64,), (128,), (64, 32), (128, 64), (128, 64, 32)\}$: Network architecture.
- $\alpha \in \{0.0001, 0.001, 0.01\}$: L2 regularization strength.
- `learning_rate_init` $(\eta) \in \{0.001, 0.01, 0.1\}$: Initial gradient descent step size.
- `max_iter` $\in \{100, 200\}$: Maximum training epochs alongside early stopping.

E. K-Means Clustering

K-Means clustering was applied to the scaled training features (besides outcome) to discover natural risk segments within the policyholder population. K-Means partitions n samples into k clusters by minimizing the within-cluster sum of squared distances:

$$\arg\min_{S} \sum_{j=1}^{k} \sum_{x \in S_j} \|x - \mu_j\|^2 \tag{10}$$

where S_j is the set of points assigned to cluster j and $\mu_j \in \mathbb{R}^d$ is the centroid of cluster j. The optimal number of clusters k was determined using two complementary methods. First, the elbow method plotted inertia (total within-cluster sum of squared distances) against k from 2 to 10, revealing diminishing returns in inertia reduction beyond $k = 3$. Second, the silhouette score:

$$s(x) = \frac{b(x) - a(x)}{\max\{a(x), b(x)\}} \tag{11}$$

where $a(x)$ is the mean distance from sample x to all other points in its cluster (intra-cluster cohesion) and $b(x)$ is the mean distance from x to all points in the nearest neighboring cluster (inter-cluster separation), was computed for each k. A silhouette score $s(x) \in [-1, 1]$ closer to 1 indicates well-separated, compact clusters. Although the elbow method suggested a soft inflection around $k = 3$, the silhouette score provided a more objective criterion, peaking at $k = 2$ (score = 0.189); therefore, selected as the final configuration, corresponding to interpretable high-risk and low-risk policyholder segments.

IV. Experiments & Results

A. Hyperparameter Tuning Results

RandomizedSearchCV identified the following optimal configurations:

- **Logistic Regression:** $C = 10$, solver = `liblinear`, `max_iter` = 500 (CV ROC-AUC: 0.902)
- **Random Forest:** `n_estimators` = 200, `max_depth` = 10, `min_samples_split` = 10, `min_samples_leaf` = 2 (CV ROC-AUC: 0.897)
- **MLP:** `hidden_layer_sizes` = $(64,)$, α = 0.0001, `learning_rate_init` = 0.001 (CV ROC-AUC: 0.899)

All three models achieved cross-validation ROC-AUC above 0.897, indicating strong and consistent generalization across folds.

TABLE I
TEST SET PERFORMANCE FOR TUNED MODELS

Model	Accuracy	F1	ROC-AUC
Logistic Regression	0.798	0.723	**0.886**
Random Forest	0.818	**0.734**	0.875
MLP	**0.828**	0.718	0.883

B. Classification Results

Logistic Regression achieved the highest ROC-AUC of 0.886, indicating the strongest overall discriminative ability between claimants and non-claimants. Critically, it achieved the highest recall on the claim class at 0.84 meaning it correctly identified 84% of actual claimants, missing only 100 out of 627. For insurance pricing, minimizing missed claimants is the most important objective, making Logistic Regression the most practically useful model despite its lower raw accuracy.

Random Forest achieved the highest F1 score of 0.734, reflecting a better balance between precision (0.68) and recall (0.80) on the claim class. It missed 126 claimants, more than Logistic Regression but fewer than MLP.

MLP achieved the highest accuracy of 82.8% but the lowest F1 score of 0.718 among tuned models. Its claim recall of 0.70 means it missed 190 actual claimants which is the most of any model. However its precision on the claim class was highest at 0.74, meaning when it predicted a claim it was most often correct.

Fig. 3 shows ROC curves for all three tuned models. Logistic Regression and MLP are the strongest discriminators (AUC 0.886 and 0.883), with Random Forest slightly behind (AUC 0.875). All three substantially outperform the random chance baseline.

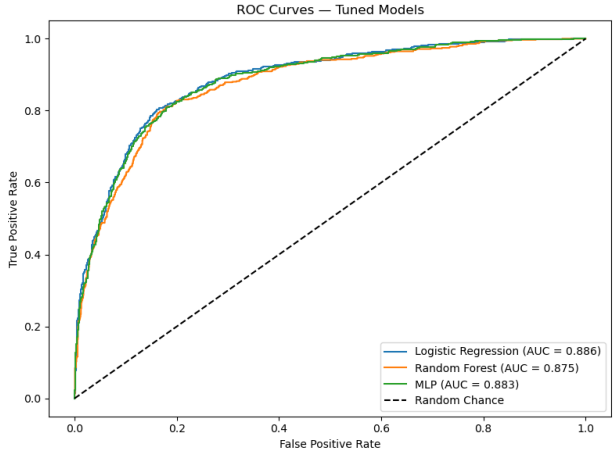

Fig. 3. ROC curves for all three tuned models.

Fig. 4 shows Precision-Recall curves. Logistic Regression achieves the highest Average Precision of 0.790, more than double the random chance baseline of 0.314.

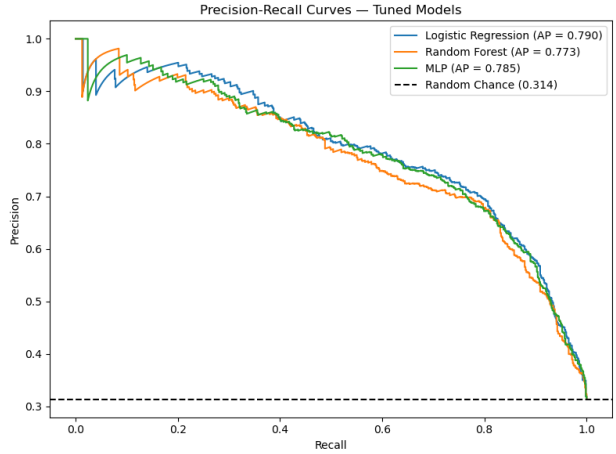

Fig. 4. Precision-Recall curves for all three tuned models. All models substantially outperform the random chance baseline (AP = 0.314).

Fig. 5 reinforces the recall tradeoff: Logistic Regression missed only 100 claimants vs 126 for Random Forest and 190 for MLP.

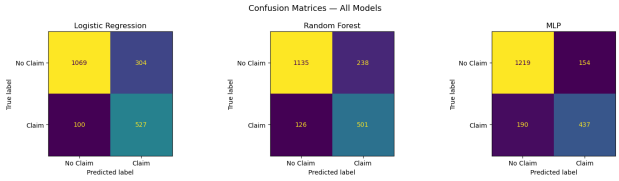

Fig. 5. Confusion matrices for all three tuned models.

Fig. 6 shows the MLP training curve. Training loss drops sharply in the first 5 epochs and stabilizes around epoch 10, while validation score plateaus near 0.83. The absence of divergence between training and validation confirms early stopping successfully prevented overfitting, terminating at epoch 19.

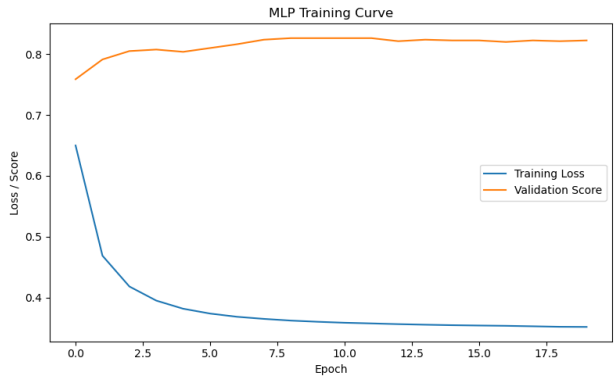

Fig. 6. MLP training loss and validation score per epoch.

C. Clustering Results

1) Optimal k Selection: The elbow method evaluated inertia across $k \in \{2, \ldots, 10\}$. The curve decreased smoothly without

a sharp bend, with a subtle inflection around $k = 3$. The silhouette score provided a more objective criterion, peaking at $k = 2$ (score = 0.189) and declining monotonically for all larger values. Based on these two complementary methods, $k = 2$ was selected.

2) Cluster Profiles: K-Means partitioned the 8,000 training samples into Cluster 0 (3,608 samples, 45.1%) and Cluster 1 (4,392 samples, 54.9%). Cluster 0 exhibited lower age (-0.782), driving experience (-0.634), income (-0.847), credit score (-0.648), vehicle ownership (-0.424), and higher annual mileage ($+0.399$), consistent with a younger, less financially stable, higher-mileage high-risk segment. Cluster 1 exhibited the opposite profile, consistent with an older, experienced, financially stable low-risk segment.

Fig. 7 shows scaled feature averages per cluster, revealing clear separation across all features.

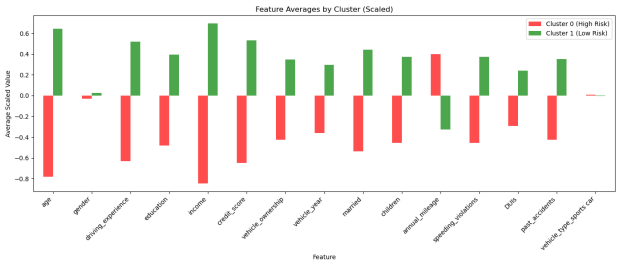

Fig. 7. Scaled feature averages by cluster. Cluster 0 (High Risk) exhibits lower age, experience, income and credit score, with higher annual mileage. Cluster 1 (Low Risk) shows the opposite profile.

3) Cluster Validation: Cluster 0 exhibited a claim rate of 55.8% while Cluster 1 exhibited only 11.2%, a nearly 5× difference. This confirms K-Means successfully discovered risk structure purely from policyholder demographics and driving history.

Fig. 8. Claim rate by cluster. K-Means discovered a 5x difference in claim rate without accessing outcome labels during training.

4) PCA Visualization: PCA compressed the 15-dimensional feature space into 2 dimensions for visual inspection. PC1 and PC2 captured 29.7% and 11.2% of total variance respectively (40.9% combined). The scatter plot shows clear spatial separation along PC1, Cluster 0 concentrated on the left, Cluster 1 on the right, with expected overlap near the cluster boundary reflecting genuinely ambiguous risk profiles.

Fig. 9 confirms spatial separation between clusters along PC1, with expected overlap near the boundary.

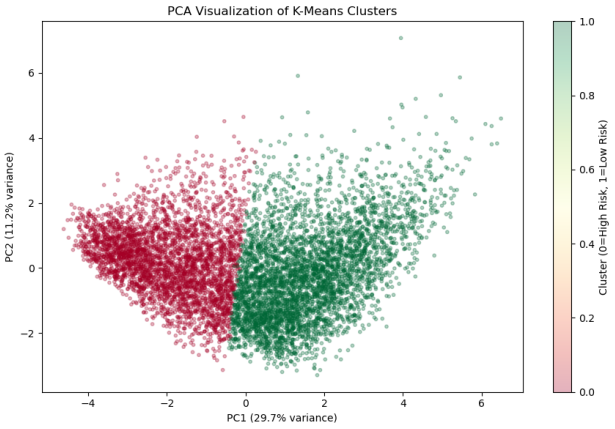

Fig. 9. PCA visualization of K-Means clusters. PC1 and PC2 explain 40.9% of total variance.

5) Cluster as Predictive Feature: Appending the cluster label as an additional feature to Logistic Regression produced negligible improvement. ROC-AUC remained at 0.886 and F1 improved only from 0.723 to 0.724. This indicates the classifier already implicitly captures the risk structure clustering made explicit. The clusters therefore serve as an interpretable summary of risk segments rather than providing novel predictive signal.

6) Feature Importance: Random Forest feature importance revealed `driving_experience` as the most predictive feature (importance = 0.234), followed by `vehicle_ownership` (0.112), `age` (0.107), `speeding_violations` (0.096), and `vehicle_year` (0.089). This directly aligns with the cluster profiles. The features most distinguishing the two risk segments are the same features the supervised model relies on most. Notably `DUIs` and `vehicle_type_sports_car` ranked lowest, suggesting frequency of violations matters more than severity of individual incidents.

Fig. 10 shows Random Forest feature importances. `driving_experience` dominates at 0.234, directly aligning with the cluster separation observed in Fig. 7.

V. Discussion & Conclusion

A. Discussion

This project was originally scoped as a two-stage supervised learning framework: a classification stage predicting claim occurrence followed by a regression stage predicting claim severity conditional on a claim occurring. However, during dataset

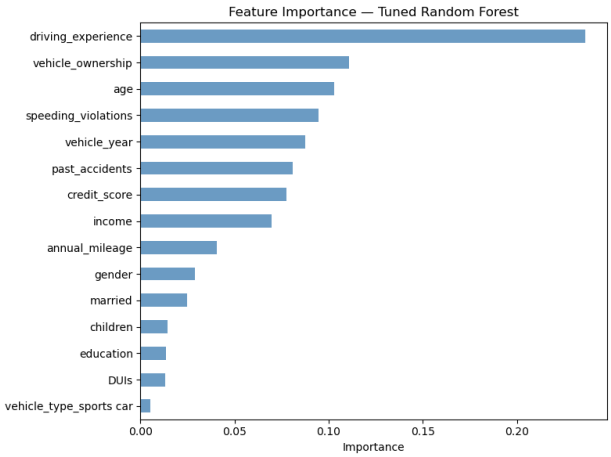

Fig. 10. Random Forest feature importance scores. `driving_experience` is the strongest predictor.

selection it became clear that no single dataset contained both non-claimants and claimants alongside a claim amount column. Datasets with a binary claim column lacked severity data, while datasets with claim amounts contained only records of policyholders who had already filed. Binary classification would be impossible as there was no negative class to learn from. Combining multiple datasets was considered but rejected, as the feature spaces were inconsistent and row counts were insufficient to produce a reliable merged dataset. The regression stage was therefore dropped and replaced with K-Means clustering, which provided a meaningful unsupervised complement to classification by discovering natural risk segments without requiring claim amount labels. Model selection was performed via 5-fold cross-validation on the training set, with the test set held out entirely for final evaluation, ensuring no data leakage between tuning and assessment.

The classification results demonstrate that policyholder demographics and driving history carry strong predictive signal for claim occurrence. Logistic Regression achieved the highest ROC-AUC of 0.886 and claim recall of 0.84, correctly identifying 84% of actual claimants. For insurers, minimizing missed claimants is more costly than false alarms, making recall the most practically important metric. The simplest model performing best suggests the underlying decision boundary is approximately linear, as complex models like MLP gained accuracy at the cost of missing more claimants.

The MLP training curve showed clean convergence with no divergence between training loss and validation score, confirming early stopping effectively prevented overfitting.

For Random Forest, overfitting was mitigated through the tuned `max_depth = 10` constraint, which prevented trees from growing until all leaves were pure. Additionally, `min_samples_split = 10` and `min_samples_leaf = 2` further regularized individual trees. Ensemble averaging across 200 trees also inherently reduces variance relative to any single decision tree.

For Logistic Regression, the tuned $C = 10$ corresponds to a regularization strength of $\lambda = \frac{1}{10} = 0.1$, a relatively weak regularization, suggesting the model was not at significant risk of overfitting on this 8,000-sample training set. This is consistent with the linear nature of the decision boundary, as simpler models with fewer parameters require less regularization to generalize well.

The clustering results provided a compelling validation of the feature set's discriminative power. K-Means with $k = 2$ naturally separated policyholders into a high-risk group (55.8% claim rate) and a low-risk group (11.2% claim rate) without ever accessing the outcome label. This $5\times$ difference in claim rate between unsupervised clusters confirms that demographic and driving history features encode genuine risk information. The cluster profiles aligned directly with intuition: younger, less experienced, lower income, higher mileage policyholders in the high-risk cluster versus older, experienced, financially stable, lower mileage policyholders in the low-risk cluster.

B. Conclusion

This project presented a classification and clustering framework for auto insurance claim prediction using policyholder demographics and driving history. Among the three supervised classifiers that were trained and evaluated, Logistic Regression emerged as the most practically effective model due to its superior recall on the claim class. K-Means clustering independently discovered two risk segments that aligned strongly with actual claim behavior, validating the predictive signal embedded in the feature set.

The original two-stage regression framework was adapted due to dataset constraints, a limitation that highlights the real-world challenge of finding labeled data that supports both claim occurrence and severity modeling simultaneously. Future work could address this by constructing a synthetic combined dataset, sourcing proprietary insurer data, or reframing severity prediction as a multi-class ordinal classification problem rather than continuous regression.

Additional future directions include exploring gradient boosted models such as XGBoost or LightGBM, which typically outperform Random Forest on tabular insurance data, incorporating telematics or usage-based features for richer driving behavior modeling, and investigating fairness-aware modeling approaches to ensure predictions do not perpetuate systemic bias across demographic groups.

References

[1] S. Bhoji, T. N. Pandey, B. B. Dash, R. Satpathy, U. C. De, and S. S. Patra, "Performance Analysis of Classifiers in Predicting Car Insurance Claim," in *Proc. IEEE Conf.*, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10677634

[2] R. Cholsan, "Customer Data," *Kaggle*, 2024. [Online]. Available: https://www.kaggle.com/datasets/racholsan/customer-data